Confidence-Aware View Planning for High-Quality Aerial 3D Reconstruction

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1 Problem Statement

High-quality aerial 3D reconstruction depends strongly on where images are captured. In urban scenes, a simple grid or overhead trajectory may miss walls, concave structures, and occluded regions, producing incomplete geometry or weak novel-view rendering quality. A practical solution is a two-pass pipeline: first build a coarse model from an initial flight, then use the coarse geometry to plan a second-pass capture path.

The problem addressed in this project is how to allocate a fixed second-pass view budget to improve reconstruction quality. In many planning pipelines, surface samples on a coarse model are marked as scanned when they are covered by a selected view. However, two observations are not equally useful: a view that is too far, too oblique, or weakly aligned with the surface normal may contribute less reconstruction information than a well-positioned view. Therefore, a binary scanned-count criterion may overestimate the true observation quality of difficult regions.

This problem is closely related to Geometric Modeling because the planning process is driven by a reconstructed geometric representation: point clouds, surface samples, normals, visibility, view rays, and camera-surface distances. The proposed method analyzes these geometric relations to guide image acquisition. The goal is not to build a completely new reconstruction system, but to study whether a confidence-aware modification to sample-based view planning can improve rendering quality.

The main technical challenges are: (1) estimating useful coverage from a noisy coarse reconstruction, (2) selecting useful second-pass viewpoints from geometric confidence cues, and (3) evaluating whether the planning-level change leads to better 3DGS rendering metrics such as PSNR, SSIM, and LPIPS.

2 Related Work

Yan *et al.* proposed a sampling-based path planning method for high-quality aerial 3D reconstruction of urban scenes [1]. Their method follows an explore-and-exploit strategy. First, an initial flight is used to reconstruct a coarse point cloud. Then, the coarse model is analyzed using Poisson-surface-based quality evaluation. Low-quality samples are assigned local viewpoints, while the remaining surface samples are covered by global viewpoints. Finally, the selected viewpoints are connected into a smooth trajectory using an ant-colony-optimization formulation.

A key part of their viewpoint selection is the confidence between a candidate viewpoint v_i and a surface sample s_j:

$$c(v_i, s_j) = w_v(v_i, s_j)\, c_d(v_i, s_j)\, c_o(v_i, s_j), \tag{1}$$

where w_v is visibility, c_d measures distance suitability, and c_o measures orientation suitability. This confidence is used to evaluate candidate views. However, after local viewpoints are chosen, the method conceptually updates a scanned mark for nearby samples

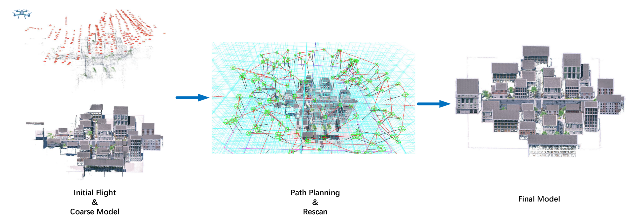

Figure 1: Overview of the two-pass UAV path planning pipeline of Yan *et al.* [1]. A coarse model is reconstructed from the initial flight, then a second-pass path is planned to rescan the scene. Figure adapted from Yan *et al.* [1].

and later selects global viewpoints for samples whose scanned mark is smaller than two. This binary coverage rule treats all covered observations as equally useful during coverage accounting, even though the paper already defines a viewpoint-sample confidence score for evaluating individual candidate views.

This project adopts the same high-level idea but studies it under a controlled budgeted setting. Instead of asking only how many times a sample has been covered, we ask how much effective observation confidence it has received. The proposed modification does not replace the original surface sampling or viewpoint confidence design; it changes how a fixed number of second-pass viewpoints is allocated.

Regarding availability, Yan *et al.* do not appear to provide a complete official source-code release for the full path-planning pipeline. The paper reports using COLMAP for reconstruction and mentions released files from the benchmark of Smith *et al.* , such as planned trajectories, captured images, and proxy models, but these are comparison/benchmark resources rather than a complete implementation of Yan *et al.* 's method. Therefore, this project uses our own controlled Yan-style reimplementation as the baseline. The goal is a controlled ablation within the same codebase, not an exact reproduction of every engineering detail in the original system.

The final evaluation will use 3D Gaussian Splatting (3DGS) [2] as a rendering-based reconstruction backend. In this setting, view selection affects the training-view distribution and can be evaluated using held-out novel views. COLMAP [3, 4] may be used for camera poses and sparse/dense reconstruction when needed.

3 Methodology

3.1 Budget-Controlled Yan-Style Baseline

This project formulates the experiment as a budgeted second-pass view selection problem. Both methods start from the same first-pass image set, the same coarse reconstruction, and the same candidate viewpoint pool. Each planner then selects exactly K second-pass camera poses. The selected poses are used to capture or render second-pass images, which are combined with the same first-pass images for downstream 3DGS training.

The baseline is a controlled Yan-style reimplementation inspired by Yan *et al.* [1]. It preserves the central idea of sample-based planning from a coarse model, but it is not intended to exactly reproduce every low-level engineering detail of the original system, since a complete official implementation is not available. For the controlled experiment, the baseline selects the K second-pass views using binary scanned-count reduction. Let $B(s)$ be the number of selected views that cover sample s. The remaining binary deficit is $D_{base}(s) = \max(0, B_{req} - B(s))$. A candidate view is preferred if it covers many samples whose binary scanned count is still insufficient:

$$G_{base}(v) = \sum_{s \in S} \min\big(D_{base}(s),\ \mathbf{1}[v \text{ covers } s]\big), \tag{2}$$

where $B_{req} = 2$ represents the common geometric requirement that a surface sample should be observed by multiple views. This baseline treats each covered observation as one count, regardless of whether the observation is strong or weak.

3.2 Confidence-Weighted Coverage Deficit

The proposed method does not introduce a new viewpoint-sample confidence function. Instead, it reuses the confidence score defined by Yan et al., which combines visibility, distance suitability, and orientation suitability. The key modification is how this confidence is used during fixed-budget second-pass view selection. The proposed method replaces binary scanned-count reduction with confidence-weighted coverage-deficit reduction. The original viewpoint confidence between a candidate view v and a sample s is

$$c(v, s) = w_v(v, s)\, c_d(v, s)\, c_o(v, s), \tag{3}$$

where w_v is visibility, c_d measures distance suitability, and c_o measures orientation suitability. Invisible views contribute zero because of the visibility term. Visible views that are too far, too close, or highly oblique contribute less than well-positioned views.

While the baseline stores how many times a sample has been covered, the proposed method stores how much effective observation confidence the sample has accumulated. Let V_{sel} be the set of already selected second-pass viewpoints. For each surface sample s, we define its accumulated effective coverage as

$$H(s) = \sum_{v \in V_{sel}} c(v, s). \tag{4}$$

Instead of asking only whether a sample has been covered twice, the proposed method asks whether the sample has received enough effective observation confidence. We define the remaining coverage deficit as

$$D(s) = \max(0, H_{req} - H(s)), \tag{5}$$

where $H_{req} = 2$ corresponds to the ideal target of two high-confidence observations. Let V_c denote the shared candidate viewpoint pool. At each greedy selection step, the proposed planner chooses the unselected candidate viewpoint that gives the largest reduction in total confidence-weighted deficit:

$$v^* = \arg\max_{v \in V_c \setminus V_{sel}} \sum_{s \in S} \min(D(s), c(v, s)). \tag{6}$$

After each greedy selection step, the selected viewpoint v^* is added to V_{sel} and removed from the candidate pool. Equivalently, the accumulated coverage in Eq. 4 is updated by adding the new contribution $c(v^*, s)$ for each surface sample s. The process repeats until exactly K second-pass viewpoints have been selected.

3.3 Optional Altitude-Aware Path Cost

As a secondary path-quality refinement, we also consider a mild altitude-aware term in the path switching cost. The original path cost combines translation and rotation:

$$C(v_i, v_j) = 0.5 C_t(v_i, v_j) + 0.5 C_r(v_i, v_j). \tag{7}$$

Although the 3D translation term implicitly includes height difference, it does not separately penalize frequent climb and descent. We therefore test

$$C'(v_i, v_j) = \alpha C_t(v_i, v_j) + \beta C_r(v_i, v_j) + \gamma C_z(v_i, v_j), \tag{8}$$

where

$$C_z(v_i, v_j) = \frac{|z_i - z_j|}{\max_{a,b} |z_a - z_b|}. \tag{9}$$

A default setting is $\alpha = 0.45$, $\beta = 0.45$, and $\gamma = 0.10$. This term is not claimed to directly improve rendering quality; it is mainly evaluated by reduced vertical motion and improved flight practicality.

3.4 Implementation Plan

The implementation will compare two variants in the same codebase: (A) a budget-controlled Yan-style baseline, and (B) the proposed confidence- and path-aware planner. All variants use the same first-pass reconstruction, the same candidate viewpoint pool, and the same second-pass view budget K.

The main code change is in the budgeted second-pass viewpoint selection stage. Instead of selecting views by binary scanned-count reduction, the proposed planner accumulates the confidence-weighted effective coverage defined in Eq. 4 and uses the coverage-deficit objective in Eq. 6 to prioritize candidate viewpoints. After the K viewpoints are selected, the proposed method also uses an altitude-aware path cost when connecting the selected viewpoints into a flight trajectory.

4 Expected Results

The primary evaluation will be based on novel-view rendering quality after training 3DGS models. Both methods start from the same first-pass image set and the same coarse reconstruction. Each planner selects exactly K second-pass camera poses from the same candidate viewpoint pool. Images are captured or rendered at these selected poses and combined with the same first-pass images to train separate 3DGS models. The models are evaluated on the same held-out views using PSNR, SSIM, and LPIPS. We expect the confidence-aware planner to improve the average PSNR and SSIM and reduce the average LPIPS over the full held-out view set, because the same number of second-pass viewpoints is allocated using effective observation confidence rather than binary scanned counts.

References

[1] F. Yan, E. Xia, Z. Li, and Z. Zhou. Sampling-Based Path Planning for High-Quality Aerial 3D Reconstruction of Urban Scenes. Remote Sensing, 13(5):989, 2021.

[2] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. ACM Transactions on Graphics, 42(4), 2023.

[3] J. L. Schoenberger and J.-M. Frahm. Structure-from-Motion Revisited. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pages 4104–4113, 2016.

[4] J. L. Schoenberger, E. Zheng, M. Pollefeys, and J.-M. Frahm. Pixelwise View Selection for Unstructured Multi-View Stereo. In Proceedings of the European Conference on Computer Vision, pages 501–518, 2016.